Short And Low Resolution Deepfake Video Detection Using CNN

1st Ashifur Rahman
dept. of CSE,BUBT
Dhaka, Bangladesh
17182103357@cse.bubt.edu.bd

2nd Nipo Siddique
dept. of CSE,BUBT
Dhaka, Bangladesh
17182103215@cse.bubt.edu.bd

3rd Mohasina Jannat Moon
dept. of CSE,BUBT
Dhaka, Bangladesh
17182103092@cse.bubt.edu.bd

4th Tahera Tasnim
dept. of CSE,BUBT
Dhaka, Bangladesh
17182103195@cse.bubt.edu.bd

5th Mazharul Islam
dept. of CSE,BUBT
Dhaka, Bangladesh
17182103205@cse.bubt.edu.bd

6th Md. Shahiduzzaman
dept. of CSE,BUBT
Dhaka, Bangladesh
shahid@bubt.edu.bd

7th Samsuddin Ahmed
dept. of CSE,BUBT
Dhaka, Bangladesh
sam@bubt.edu.bd

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***Abstract*—Recently, convincing deepfake videos are growing very fast that can delude even the trained experts. These deepfake videos have huge impacts all over the world covering the political, social, and personal lives. The state-of-the-art machine learning studies are demonstrating noticeable success to detect fake videos in high resolution and long-time video data while the same performance is not observed in low resolution and short-time clips. In this work, we have trained a convolutional neural network (CNN) that demonstrates mentionable accuracy in detecting fake videos in low-resolution and short-time video data. We have exploited Kaggle Deepfake Detection Challenge (DFDC) and the Face Forensics++ datasets in our experiment. Our model shows 94.93% accuracy in detecting fake videos for the DFDC dataset while the same is 93.2% for FaceForensics++ Dataset. We evaluated our models by different performance metrics and compared the performance with state-of-the-art methods. Our model demonstrates comparable performance.**

***Index Terms*—Deepfake Video, Machine-Learning, dlib, Convolutional Neural Network(CNN), Deepfake-Detection-Challenge(DFDC), FaceForensics++.**

I. Introduction

Deepfake videos are manipulated videos that use Machine Learning based algorithms to swap a person in an existing image or video with someone or something else [1]. Deepfake is currently one of the most important concerns confronting modern civilization [2], [3]. Deep fake videos are regularly used to slander and harm the reputation of celebrities, politicians, and other well-known personalities [4], [5]. These deep fake videos are spreading wreak havoc in the communities and countries around the globe [4].

The exponential growth of artificial intelligence and deep learning-based accessible modern tools, techniques, and software such as convolutional neural network [6], autoencoders [7], generative adversarial network [8], TensorFlow [9], Keras [10], PyTorch [11] , FaceApp [12], FakeApp [13] etc enabled image and video editing open to almost anybody with a computer [14].

Several studies are detecting deepfakes are available in the literature [14]. The majority of these studies were based on deep learning methods [15]. They are fairly accurate when the sample video is too long and the quality is high. However, their accuracy was not as good in short and low-resolution videos.

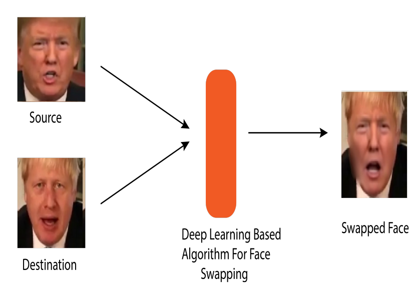

Fig. 1. Overview of DeapLearning based face-swapping system, The system take two input for image manipulation one is source image and another one is the destination image. Then deep learning-based face swapping algorithm swapped the source image face into the destination image face, and create a deepfake image. Which is the final output of this system.

In this research, we propose a deepfake video recognition approach based on CNN. Now here we summarized our work, we used dlib to detect the face of a person from a video, then we extract image frame by frame from a video, and then we use those images to train our CNN [6] model to classify if the video is real or fake. We used Kaggle Deepfake-Detection-Challenge(DFDC) [16] dataset and modified the FaceForensics++ [17] dataset to train and validate our model.

II. Related Work

In the field of deepfake video and image manipulation detection, not so much work has been done. This field is very new to a researcher. Here we review some of the past work of this field. This paper [18] has proposed a recurrent neural network (RNN) [19] based model to detect deepfake video. With the help of a convolutional neural network (CNN) [6] and long short term memory(LSTM) [20], they developed a new technique called convolutional LSTM (Conv-LSTM).They are using CNN to extract features of each frame and LSTM for video analysis. In CNN they adopted the InceptionV3 [21] as the fully connected top layer of the network. Using a pre-trained ImageNet [22] model they cropped the image into the desired size. Now the output image of CNN goes to LSTM network analysis. The frame is fake or real. They employed a 2048-by-2048 LSTM with a 0.5 dropout rate. After that, LSTM was used to join 512 fully connected layers with a 0.5 percent of dropout. At the last layer, they used softmax to compute the probabilities of the video being real or not. They utilized the Adam [23] optimizer with a learning rate of 1e-5 and decay of 1e-6 to construct the model. A total of 600 videos were utilized to train and validate this system. They gathered 300 videos from a variety of video-hosting websites, with the remaining 300 videos coming from the HOHA dataset [24]. They divided the dataset into three sections for training, validation, and testing: 70/15/15 respectively. Through this model, they will get the highest accuracy of 94%.

In the paper Deepfake Video Detection through Optical Flow-based CNN [25], they proposed a deep learning technique for the creation and processing of multimedia content. They use the PWC-Net model for optical flow, which is based on the CNN [6] model.They utilized VGG16 [26] and ResNet50 [27] as backbones to test their experiment. They used pre-trained network motion vectors that were represented as 3-channel images and then used as input for a neural network to resolve the deepfake videos challenge. They trained a generic net using randomly left-right switched rectangular patches of size 224 x224 pixels and a larger patch of 300 x300 pixels including the face. They also utilized Adam optimizer, which had a learning rate of 10-4%. They used the FaceForensics++ [17] dataset to conduct their tests, which contains 1000 actual video clips that were modified using three automated face manipulation methods: Deepfakes, Face2Face, and FaceSwap. There are 720 videos for learning, 120 for evaluation, and another 120 for the test. The accuracy of both constructed networks (VGG16 and ResNet50) is 81.61% and 75.46%, respectively, according to preliminary results.

The authors in [28] presents a technique for identifying anomalies by employing a ResNext Convolutional Neural Network (CNN) [6] and a Recurrent Neural Network (RNN) [19] with Long Short Term Memory (LSTM) [20] to compare the produced face areas and their neighboring areas. They used ResNext CNN for level detection and RNN along with LSTM for video detection. Their method is inspired by how deepfakes are created by GAN [29] with the help of Autoencoders [7]. The dataset they used is split into 70% train and 30% test sets. The model consists of 50_32x4d followed by one LSTM layer. They are using 2048 LSTM units with a 0.4 chance of dropout. Their method has not considered the audio so they will not be able to detect the audio deepfake.

In [30] the authors presented a novel Deepfake algorithm that allows a user to replace one person's face in a video with the face of another. The average standardized cross-correlation test between real videos and Deepfakes differs significantly, according to the PRNU research. The dataset features authentic, neutral video, made by Canon Powershot SX210IS and has a resolution of 1240 × 720 pixels. Powershot SX210IS and Deepfeke GUI use OpenFeedShop for Deepfake. The videos are created in a series of frames as PNG using FFPpeig's software. The frames are subdivided into eight groups of equal size and an average PRNU pattern. Created for each group using the second-order (FSTV) method with the software 'PRNU comper' software. These eight PRNU patterns are then compared to one another. Other, and sequence correlation results have been returned.

III. Methodology

In this section the proposed method,whose system architecture and system design is given Figure-2 and Figure-3 respectively.

In our deepfake detection work, at first, we process our whole train dataset video to crop frame by frame facial image. To crop facial images we detect the given video's face using dlib [31], [32] module of python. After cropping all the videos we get our process facial image dataset. Then we train our model through the dataset. We train three models to compare with them, the first one is InceptionResNetV2 [33], the second one is MobileNet [34] and the last one is DenseNet121 [35].

Respectively in all three models(InceptionResNetV2 [33], MobileNet [34], and DenseNet121 [35]), we used pre-trained ImageNet [22] weights at the first layers of every model. After that, we add a GlobalAveragePooling2D layer followed by a dense layer. We used softmax as an activation function of all three models. To compile all three models we used adam [23] optimizer with a learning rate of 1e-5 and decay of 0.0, loss function as "binary-cross-entropy" and matrix as "accuracy".

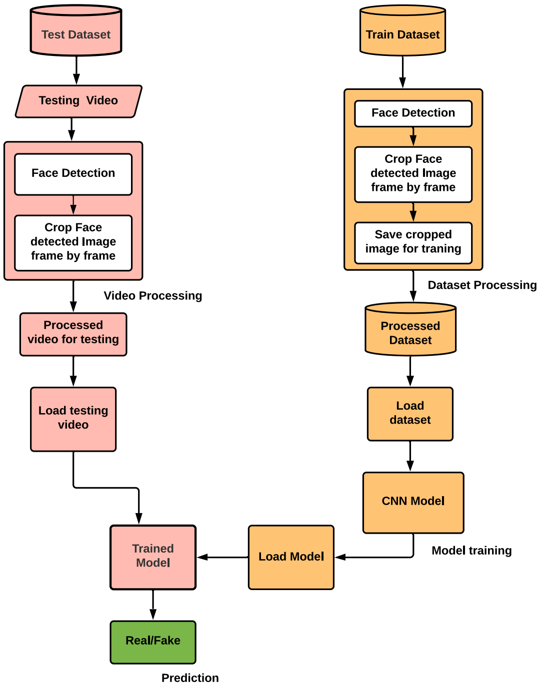

Fig. 2. System architecture, To train the model at first it detects the person's face from videos, after that it crops them frame by frame. The cropped facial image is then saved in order to train the CNN model. After finishing training now the model are ready to test the video to detect the video is fake or real. For testing a video firstly the model detects the person's face from videos and crop them frame by frame. Then the CNN model takes the cropped image to make a decision the image is fake or real. After checking all the frames the model gives the final that the testing video is real or fake.

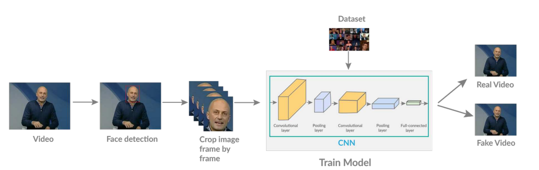

Fig. 3. Overview of our deepfake detection system,The system learns and infers from a specific(DFDC) dataset.It has a convolutional neural network to make decision the video is real or fake.

To train and validate our model we use the Kaggle Deepfake-Detection-Challenge(DFDC) [16] dataset, which consists of a total of 800 videos. We split our dataset into 70/20/10 for training, validation, and test respectively. We also use the modified FaceForensics++ [17] dataset to compare the result with our Deepfake-Detection-Challenge(DFDC) [16] dataset. In this modified FaceForensics++ [17] dataset we use 100 fake and 100 real videos to train our model and for testing, we use 15 fake and 15 real videos. Both datasets give similar results in our model.

IV. EXPERIMENTAL SETUP

In our experiment, first of all, we need a minimum of windows-7 or more as an operating system and google colab or jupyter notebook as an integrated development environment (IDE).To execute our experiment we need some python modules to install first.Those are CV2, dlib, tensorflow, sklearn, keras and matplotlib.We can easily install that module using pip install module_name command in windows command prompt for jupyter notebook and in google colab we simply run the command into a cell to install it. We use Deepfake-Detection-Challenge(DFDC) [16] dataset in our experiment. In jupyter notebook, we need to give an exact location of our dataset, which is present in our computer's local drive. To use google colab at first we need to upload the whole dataset into our google drive. Then mount the drive by google colab and give the dataset location that is present in google drive.

V. RESULT

A. EVALUATION MATRIX

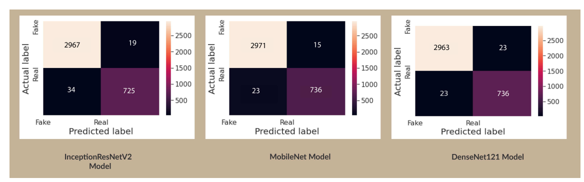

Fig. 4. Confusion Matrix for InceptionResNetV2 model & CNN model.The anticipated label is on the x-axis, and the actual label is on the y axis in this matrix. In this matrix, we mapped Real-Real as true positive(TP), Fake-Fake as true negative(TN), Fake-Real as false positive(FP), and Real-Fake as false negative(FN).

A confusion matrix is a table that shows how well a classification model (or "classifier") performed on a set of test data for which the true values are known.In confusion matrix there are only two

possible predicted classes: on is "true/yes" and the other one is "false/no".In our model we used Real or Fake instead of yes or no.In our model, "real" means the video is genuine, whereas "fake" means the video is not genuine. For the purposes of this calculation, we will refer to true positives as TP, true negatives as TN, false positives as FP, and false negatives as FN.

Accuracy: The successfully predicted divided by the total number of predictions is how accurate a model is.This is also known as accuracy.Equation of accuracy is,

$$Accuracy = (TP + TN)/total \tag{1}$$

Precision: Precision means the ratio of total Positive samples correctly classified to the total number of samples classified as Positive.Equation of precision is.

$$Precision = TP/(TP + FP) \tag{2}$$

Recall: The total number of Positive samples accurately categorized as Positive divided by the total number of Positive samples is known as recall.Equation of recall is,

$$Rcall = TP/(TP + FN) \tag{3}$$

B. RESULT ANALYSIS

It is so hard to discover deepfake recordings where the manipulation is just present in a little part of the video. To represent this, we take fixed-length images from each video in the training, validation, and test sections to provide input to our model. Figure-4 shows confusion matrix of both InceptionResNetV2, MobileNet [34] and DenseNet model's.

In figure-5 , figure-6 and figure-7 we attached both of our InceptionResNetV2 [33], MobileNet [34] and DenseNet [35] model's receiver operating characteristic (ROC) curve that are showing our model's performance in DFDC dataset.

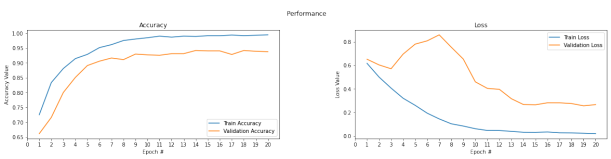

Fig. 5. InceptionResNetV2 model ROC figure.The true positive rate (TPR) is plotted on the y axis, while the false positive rate (FPR) is plotted on the x axis in this graph.To train model we take 20 epoch that are in x axis and in y axis we take Accuracy value(for a) and Loss value(for b).

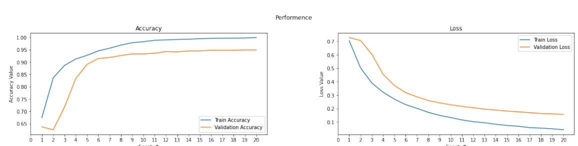

Fig. 6. MobileNet model ROC figure.The true positive rate (TPR) is plotted on the y axis, while the false positive rate (FPR) is plotted on the x axis in this graph.To train model we take 20 epoch that are in x axis and in y axis we take Accuracy value(for a) and Loss value(for b).

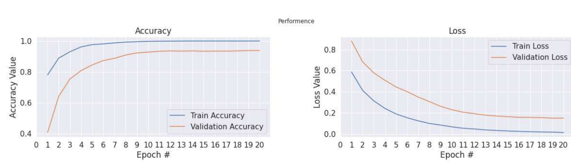

Fig. 7. DenseNet model ROC figure.The true positive rate (TPR) is plotted on the y axis, while the false positive rate (FPR) is plotted on the x axis in this graph.To train model we take 20 epoch that are in x axis and in y axis we take Accuracy value(for a) and Loss value(for b).

In the table-I, we give our model's accuracy, precison, and recall for the DFDC dataset. Our all models give high accuracy in the DFDC dataset. The accuracy of the given three models is 93.7%,94.93%, and 93.86% respectively. With the basis of accuracy, we can say that MobileNet performed better than the other two models to detect deepfake video.

TABLE I
ACCURACY TABLE FOR INCEPTIONRESNETV2, MOBILENET AND DENSENET121 CNN MODEL.HERE WE SAW ACCURACY, PRECISION, AND RECALL OF BOTH MODELS.

System Name	Accuracy	Precision	Recall
InceptionResNetV2	93.7%	98.0%	97.0%
MobileNet	94.93%	99.0%	98.0%
DenseNet121	93.86%	98.0%	98.0%

C. COMPARISON WITH OTHER METHOD

The comparison between the previously used method and our method is shown in table II.In the previously done method at first, we saw Deepfake Video Detection Using Convolutional Vision Transformer [36] uses CNN & GAN model to classify deepfake video but their accuracy is not so high. They got only 91.5% accuracy. Second, on is Deepfake Video Detection through Optical Flow-based CNN [25] which uses the CNN model and VGG16 as architecture and they get 93.0% accuracy. In this model, VGG16 has a total of 138,357,544 parameters that are so high.

In the third on we saw Deepfake Video Detection Using Recurrent Neural Networks [18] . They use both CNN and RNN to build a model which is very complicated to implement. In the fourth on we saw Video Face Manipulation Detection Through Ensemble of CNNs [5] which uses CNN as model and XceptionNet as architecture. In this method, they get only 90.0% as accuracy.

In our method, we used InceptionResNetV2 [33], MobileNet [34], and DenseNet [35] models to classify deepfake video.InceptionResNetV2, MobileNet, and DenseNet models have total 55,873,736, 4,253,864 and 8,062,504 parameters respectively, which much lower then VGG16 & XceptionNet. Although the total parameters of our model are low, we got much higher accuracy than other methods. We got 93.7% accuracy in InceptionResNetV2 [33], 94.93% accuracy in MobileNet [34] and 93.86% DenseNet [35] respectively.

VI. CONCLUSION

We have presented a temporary attentive framework in this paper to identify deepfake videos. Using a large number of controlled videos, we found that using a basic Convolutional Neural Network (CNN) [6] can accurately predict whether or not a video has been manipulated. We believe that our work provides an excellent initial line of defense against fake media created with the devices shown in the article. We demonstrate how using a simple

TABLE II
COMPARISON WITH OTHER METHODS. HERE WE COMPARING OTHER METHODS WITH OUR NEW METHOD ON THE BASIS OF THE DATASET, MODEL, ARCHITECTURE, AND ACCURACY

	System Name	Dataset	Model	Arch	Accuracy
Others Methods	Deepfake Video Detection Using Convolutional Vision Transformer [36]	DFDC	CNN & GAN	CViT	91.5%
	Deepfake Video Detection through Optical Flow based CNN [25]	FF++	CNN	VGG16	93.0%
	Video Face Manipulation Detection Through Ensemble of CNNs [5]	DFDC & FF++	CNN	XceptionNet	90.0%
Our Model	Deepfake Video Detection Using CNN	DFDC	CNN	InceptionResNetV2	93.7%
				MobileNet	94.93%
				DenseNet121	93.86%

deep learning model can achieve significant results to detect fake videos.

In the future, we want to evaluate how we may improve our system's robustness against altered videos by using unknown techniques during training.

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